



VLSI Testing

積體電路測試

Logic Built-In Self Test (BIST)

Part 1: Test Pattern Generator

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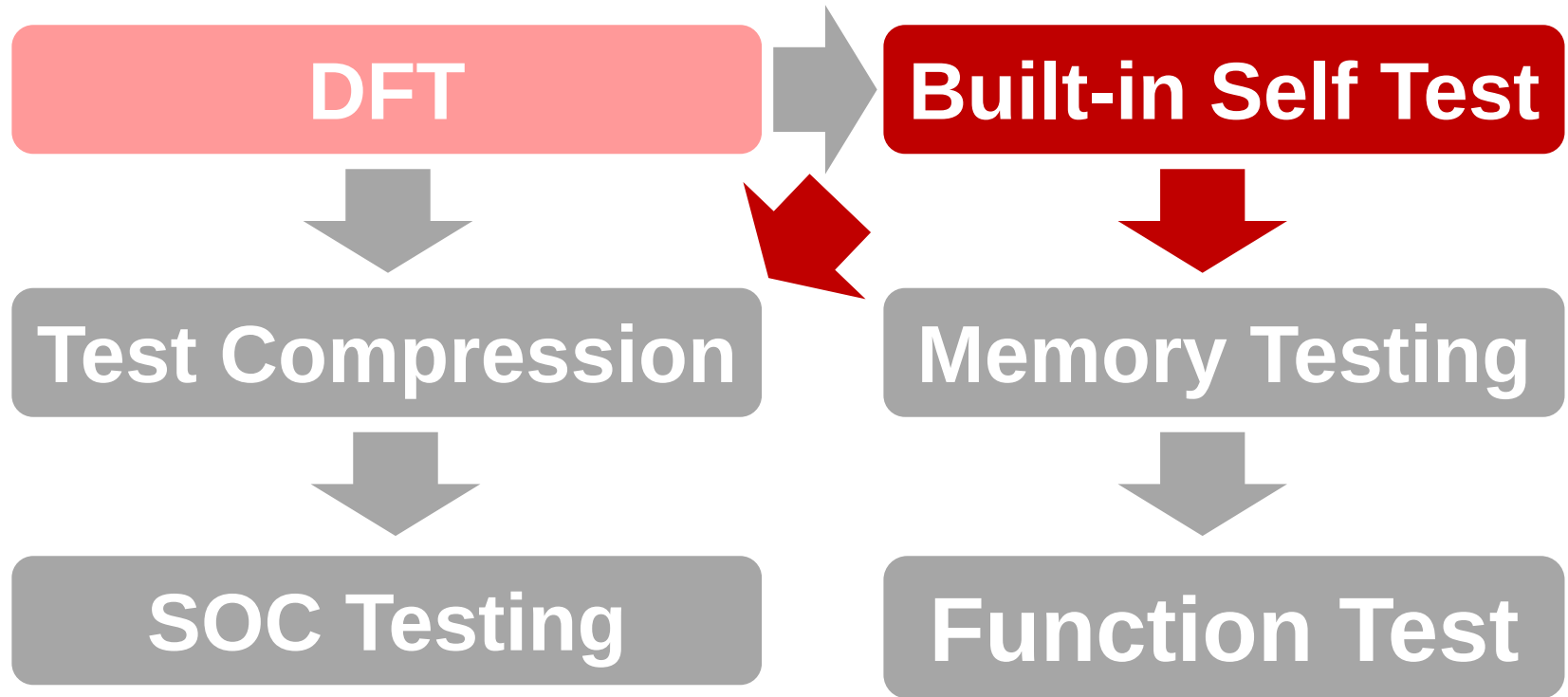
Graduate Institute of Electronics Engineering

National Taiwan University

* Some pictures are courtesy of Prof. Shi-yu Huang, NTHU

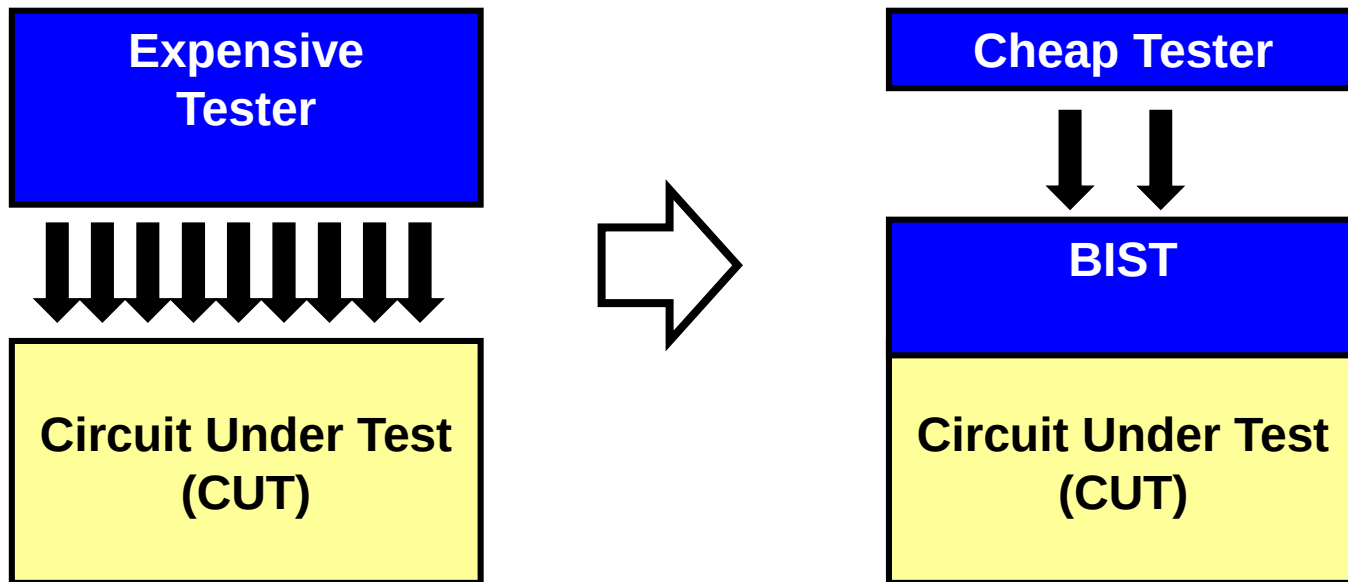
VLSI Test 13.1 © National Taiwan University

Course Roadmap (Design Topics)



Motivating Problem

- Your manager complains tester is too expensive.
- Please design a circuit to test the chip itself!



Why Am I Learning This?

- Built-in Self Test (BIST) can
 - ◆ Remove large expensive testers
 - ◆ Test chips at high speed
 - ◆ Test chips online

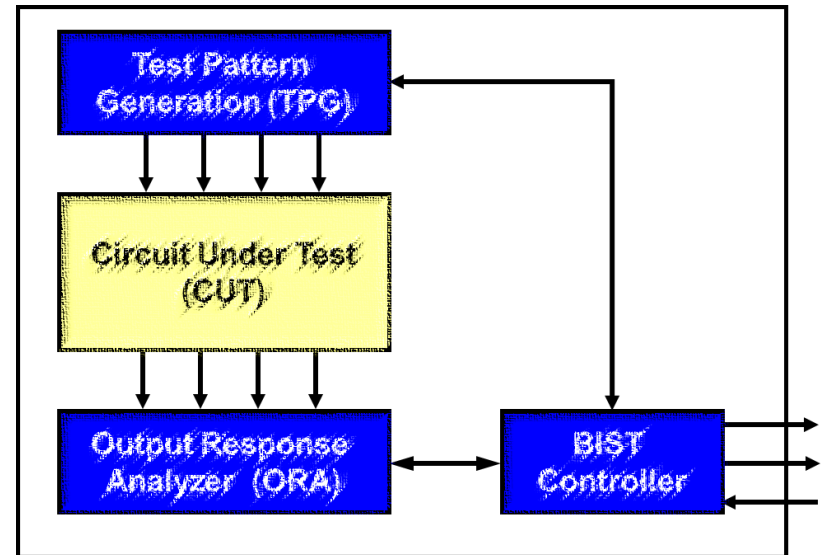
“Testing oneself is best when done alone.”

君子慎其獨也

(Jimmy Carter)

BIST Outline

- Part 1
 - ◆ Introduction
 - ◆ Test Pattern Generation
- Part 2 (Next chapter)
 - ◆ Output Response Analysis
 - ◆ BIST Architecture
 - ◆ Problems and solutions
 - ◆ Conclusions



Built-in Self Test

- Definition
 - ◆ Capability of hardware/software to carry out explicit test of itself
- Levels of BIST
 - ◆ System-level self test
 - * system self test of mainframe computer
 - ◆ Board-level self test
 - * so we can replace a bad board in a system
 - ◆ Chip-level self test
 - * focus of this lecture



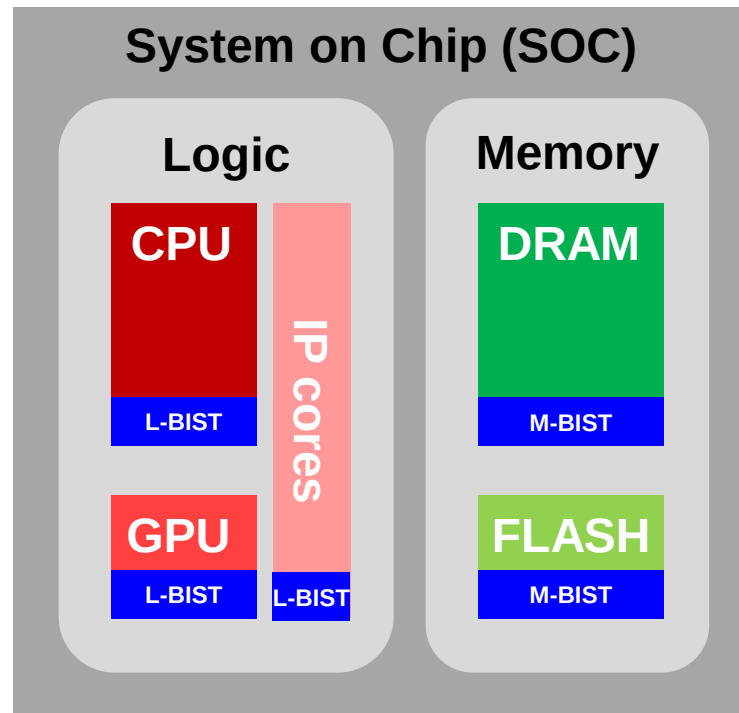
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IBM Blue Gene

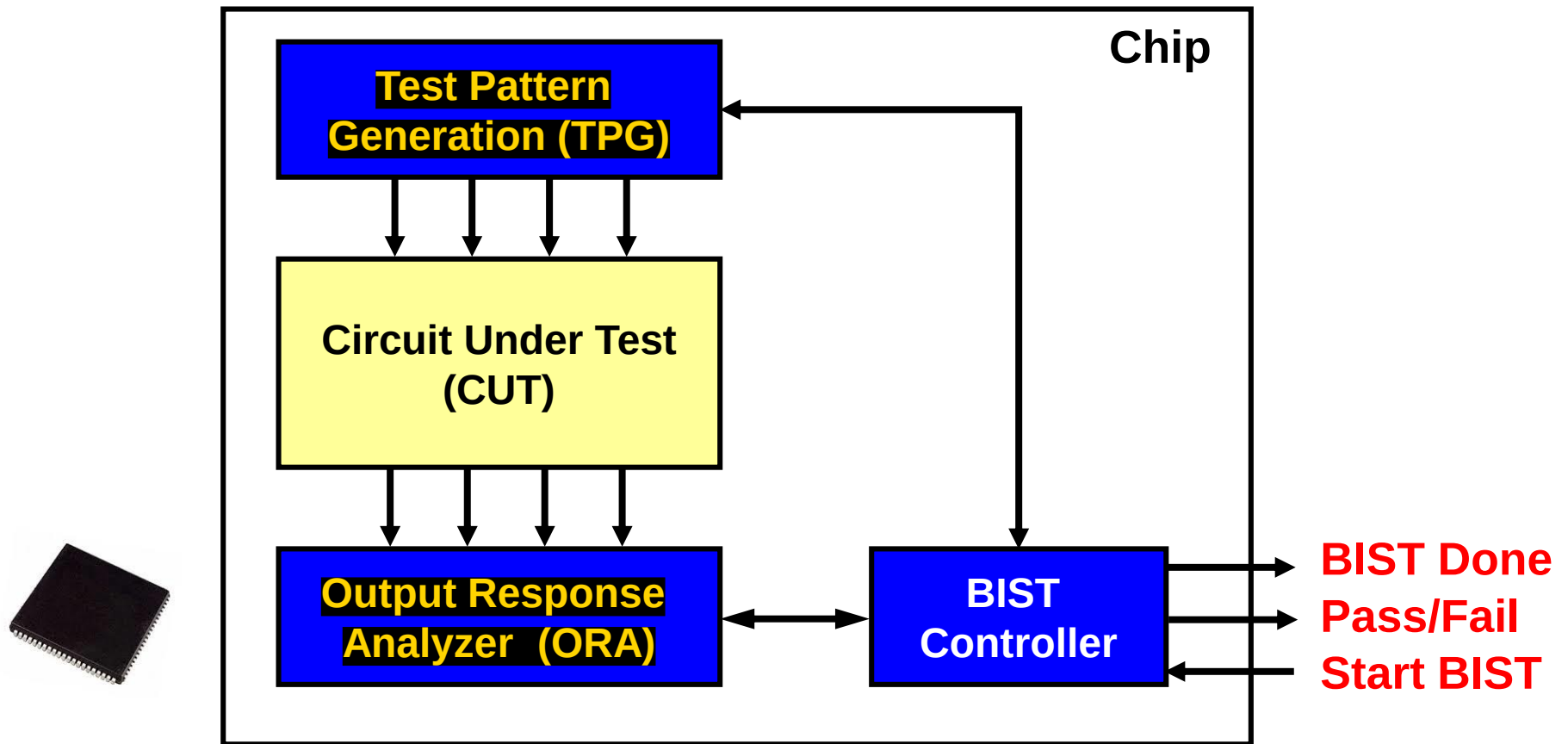
Categories of Chip-level BIST

- Based time of test
 - ♦ **On-line BIST**
 - * On-line testing while chip in normal operation
 - * e.g. error detection and correction for RAM
 - ♦ **Off-line BIST**
 - * Off-line testing while chip not in normal operation
 - * Focus of this talk
- Based on CUT
 - ♦ **Logic BIST**
 - ♦ **Memory BIST**



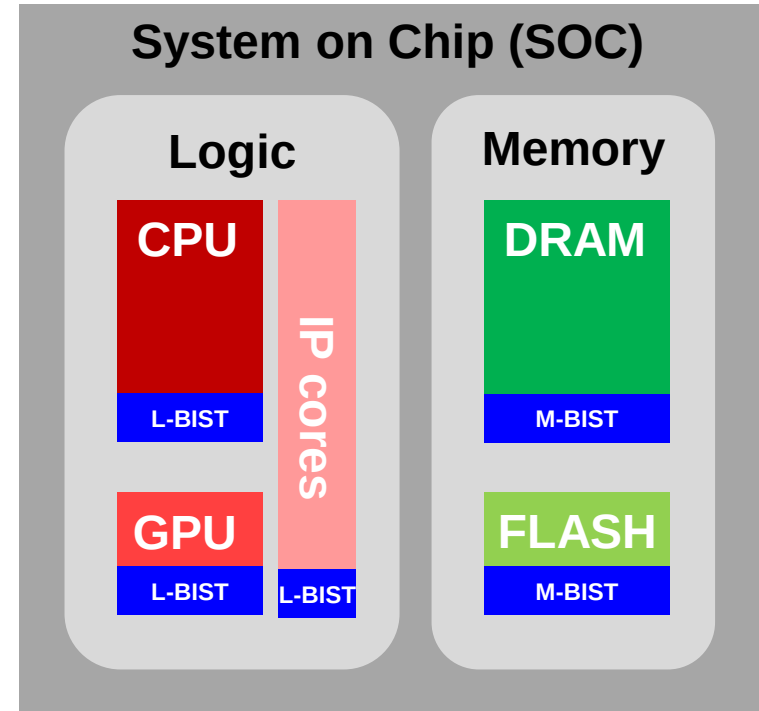
Architecture of BIST

- Three components: **BIST Controller, TPG, ORA**
- Three I/O Pins: **Start BIST, BIST Done, Pass/Fail**



Why BIST?

- 1. Save ATE cost
 - ♦ Smaller test pattern storage
 - ♦ Fewer DFT pins
 - ♦ Slower tester speed
- 2. Better IC quality
 - ♦ Test speed higher than ATE
- 3. Easier integration of tests
 - ♦ **Intellectual Property (IP)** cores
- 4. Easier test access
 - ♦ Test embedded memory in SOC
- 5. Enable on-line testing
 - ♦ Ensure reliability



BIST Has Many Unique Advantages

ATE Cost w/wo BIST

- Consider a 1 GHz μ P with 1000 pins.
- Test w/o BIST
 - ♦ 1 GHz ATE: $\$3,000/\text{pin} \times 1,000\text{pins} = \$3,000,000 \text{ USD}$
 - ♦ Huge initial capital cost
- Test w/ BIST
 - ♦ 20 MHz ATE: $\$400/\text{pin} \times 1,000\text{pins} = \$400,000 \text{ USD}$
 - * Test I/O pad contact
 - * Provide test commands, and read out test results



BIST Reduced ATE Cost

Disadvantages of BIST

- 1. Area Overhead
 - ♦ Yield loss due to BIST circuitry
- 2. Performance degradation
 - ♦ Extra hardware
- 3. Extra design effort
 - ♦ Test point insertion, BIST insertion, verification ...
- 4. Lack information for debug and diagnosis
 - ♦ Need to bypass BIST when diagnosis
- 5. Long test length but fault coverage may not good enough
 - ♦ Random test patterns not as good as ATPG patterns
 - ♦ Mixed solution (**BIST + ATE**) is often needed

BIST can NOT Solve All Problems

Quiz

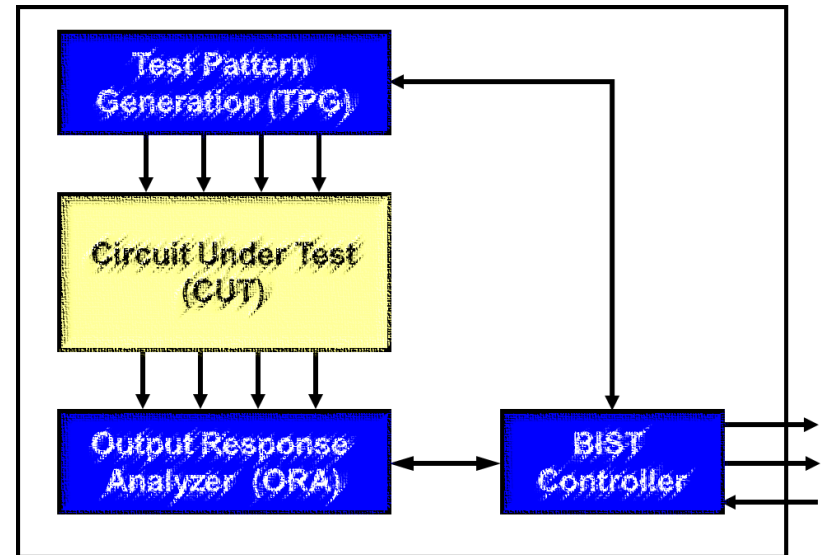
Q: Which of the following is NOT true for BIST?

- A. BIST reduces test cost because of good fault coverage**
- B. BIST reduces DFT pins required**
- C. BIST improve test quality because test speed is higher**

ANS:

BIST Part1 - TPG

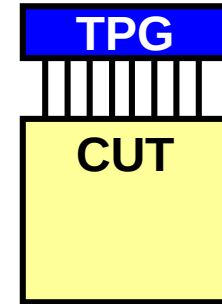
- Introduction
- Test Pattern Generation (TPG)
 - ◆ Deterministic: ROM, Algorithm, Counter
 - ◆ Pseudo Random:
 - * Linear Feedback Shift Register (LFSR)
 - * Cellular Automata (CA)



ROM/Algorithm as TPG

- **ROM as TPG**

- ♦ Store test patterns in ROM
- ♦ Very expensive for chip-level BIST
- ♦ Maybe doable for system-level BIST
 - * e.g. self test program in BIOS (Basic Input/Output System)



- **Algorithm as TPG**

- ♦ Test pattern generation based on certain mathematical rule
- ♦ Suitable for regular structure like memory, FPGA
 - * Not very useful for random logic

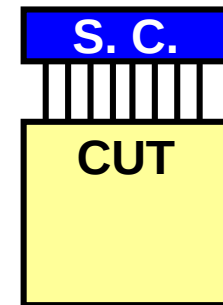
No Good for Logic BIST

Shift Counters as TPG

- Generates regular test sequences
 - ♦ Such as walking sequence for interconnect testing
- Advantage: **Linear test time**
- Disadvantage: Too regular, not useful for random logic

Walking Sequence								
cycle								
1	1	0	0	0	0	0	0	0
2	0	1	0	0	0	0	0	0
3	0	0	1	0	0	0	0	0
4	0	0	0	1	0	0	0	0
5	0	0	0	0	1	0	0	0
6	0	0	0	0	0	1	0	0
7	0	0	0	0	0	0	1	0
8	0	0	0	0	0	0	0	1

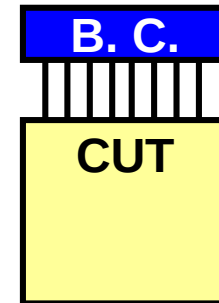
	1	2	3	4	5	6	7	8
line	1	2	3	4	5	6	7	8



← Bridging faults between lines
can be detected

Binary Counters as TPG

- N-stage binary counters
 - ♦ Generate test patterns in sequence
 - * from 0 to 2^N-1 , or from 2^N-1 to 0
- Advantage
 - + Simple design
 - + Exhaustive test is high quality
- Disadvantages
 - Exhaustive test length is very long
 - ⇒ No randomness, deterministic pattern sequence
 - ⇒ e.g. Need 2^{N-1} test length to reach a 1 at MSB
 - Large area overhead (compared to LFSR)



Need Random Counter for Logic BIST

Summary

- Introduction
 - ◆ BSIT components: **TPG, ORA, and controller**
 - ◆ Pros: **reduced pins, tester cost, on-line testing**
 - ◆ Cons: **area overhead, performance degradation, lower FC**
- Test Pattern Generation
 - ◆ Deterministic: **ROM, Algorithm, Counter**

